

# Lab07\_PedroLacerda

Pedro da Costa Lacerda

1)

```
library(RSQLite)

conn <- dbConnect(SQLite(), "database.sqlite3")
```

2)

```
query_professores <- "
  SELECT DISTINCT i.id, i.name
  FROM instructors i
  JOIN teachings t ON i.id = t.instructor_id
  JOIN sections sec ON t.section_uuid = sec.uuid
  JOIN course_offerings co ON sec.course_offering_uuid = co.uuid
  JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
  JOIN subjects s ON sm.subject_code = s.code
  WHERE s.abbreviation = 'STAT'
  ORDER BY i.name
"

professores_stat <- dbGetQuery(conn, query_professores)
cat("Quantidade de professores:", nrow(professores_stat), "\n")
```

Quantidade de professores: 379

```
print(professores_stat)
```

|   | id      | name                 |
|---|---------|----------------------|
| 1 | 4787911 | ABIGAIL BENZINE      |
| 2 | 3111842 | ADIN-CRISTIAN ANDREI |
| 3 | 3775809 | AKICHIKA OZEKI       |

|    |         |                           |
|----|---------|---------------------------|
| 4  | 4958796 | ALAN HUANG                |
| 5  | 5451706 | ALBERTO DEL PIA           |
| 6  | 3661782 | ALBRECHT KLEMM            |
| 7  | 5656034 | ALEXANDER COVINGTON       |
| 8  | 4680082 | ALEXANDER FISH            |
| 9  | 3573775 | ALEXANDER KISELEV         |
| 10 | 5162339 | ALYSSA DIGILIO            |
| 11 | 129415  | ANATOLE BECK              |
| 12 | 3674168 | ANDREJ ZLATOS             |
| 13 | 5003565 | ANDREW LESLIE             |
| 14 | 4952769 | ANNE BRUCKNER             |
| 15 | 5498092 | ANQI SHI                  |
| 16 | 4646729 | ANRU ZHANG                |
| 17 | 2600893 | ARNOLD MILLER             |
| 18 | 4963378 | AUGUST JENSEN             |
| 19 | 5852265 | BEHZAD AALIPUR            |
| 20 | 5694563 | BEHZAD AALIPUR HAFSHEJANI |
| 21 | 4140469 | BEN ADAM HAALAND          |
| 22 | 4538519 | BENEDEK VALKO             |
| 23 | 4632211 | BENJAMIN RECHT            |
| 24 | 5906523 | BI CHENG WU               |
| 25 | 4155990 | BIN DAI                   |
| 26 | 5186047 | BIN ZHANG                 |
| 27 | 4436066 | BIN ZHU                   |
| 28 | 5144884 | BINGYING XIE              |
| 29 | 3484236 | BO HUANG                  |
| 30 | 5632570 | BO YANG                   |
| 31 | 5690247 | BOWEN HU                  |
| 32 | 4753018 | BRET HANLON               |
| 33 | 751075  | BRET LARGET               |
| 34 | 2602055 | BRIAN YANDELL             |
| 35 | 5011312 | BROOK LUERS               |
| 36 | 3955546 | CECILE ANE                |
| 37 | 5883206 | CHAN PARK                 |
| 38 | 3976099 | CHANHAN HSU               |
| 39 | 4724058 | CHAOQUN MEI               |
| 40 | 4712098 | CHAOYANG YU               |
| 41 | 3376532 | CHELSEY GREEN             |
| 42 | 4586558 | CHEN CHENG                |
| 43 | 5018463 | CHEN JING                 |
| 44 | 5592111 | CHENGNING ZHANG           |
| 45 | 4597046 | CHENLIANG XU              |
| 46 | 4903489 | CHENSHENG KUANG           |

|    |         |                          |
|----|---------|--------------------------|
| 47 | 3973813 | CHENXI LI                |
| 48 | 3629214 | CHIA-CHIEH LIN           |
| 49 | 3613299 | CHIEN-WEI CHEN           |
| 50 | 340544  | CHRISTINA M. KENDZIORSKI |
| 51 | 138086  | CHRISTINE SORKNESS       |
| 52 | 3855025 | CHRISTOPHER WAGNER       |
| 53 | 5887049 | CLAIRE BOBST             |
| 54 | 4721578 | CLAUDIA SOLIS LEMUS      |
| 55 | 4312759 | COLE COOK                |
| 56 | 5332940 | COLIN LONGHURST          |
| 57 | 5498648 | CRYSTAL CHEN             |
| 58 | 5015194 | CUICUI QI                |
| 59 | 5176022 | CUIZE HAN                |
| 60 | 5846795 | DANIELE CAPPELLETTI      |
| 61 | 4308515 | DAVID ANDERSON           |
| 62 | 2601596 | DAVID DEMETS             |
| 63 | 2602109 | DAVID GRIFFEATH          |
| 64 | 6043299 | DEBRAJ DAS               |
| 65 | 4264265 | DEREK BEAN               |
| 66 | 5305675 | DEREK NORTON             |
| 67 | 3773167 | DEYUAN JIANG             |
| 68 | 2600233 | DIETRICH UHLENBROCK      |
| 69 | 393380  | DONALD PORTER            |
| 70 | 4898014 | DONG XIA                 |
| 71 | 4897725 | DONGGYU KIM              |
| 72 | 5620719 | DONGHYUN LEE             |
| 73 | 2602050 | DOUGLAS M. BATES         |
| 74 | 4728844 | DUY NGUYEN               |
| 75 | 5516726 | DUZHE WANG               |
| 76 | 5189535 | EDWARD ERKER             |
| 77 | 2824494 | ELOISA D CHAVAS          |
| 78 | 3566378 | ERICA LEE DEADMAN        |
| 79 | 2602042 | ERIK NORDHEIM            |
| 80 | 5204089 | FAN GAO                  |
| 81 | 4437355 | FAN YANG                 |
| 82 | 3460189 | FANG FANG                |
| 83 | 3783301 | FANGFANG WANG            |
| 84 | 4601948 | FLORIAN BERTRAND         |
| 85 | 361804  | FREDERICK BOEHM          |
| 86 | 5214164 | GARVESH RASKUTTI         |
| 87 | 495087  | GARY HOWARD SCHROEDER    |
| 88 | 4957688 | GENG LI                  |
| 89 | 5017655 | GINA BENNINGER           |

|     |         |                               |
|-----|---------|-------------------------------|
| 90  | 5862942 | GINA OH                       |
| 91  | 5320625 | GONZALO CONTADOR              |
| 92  | 2600488 | GRACE WAHBA                   |
| 93  | 4761200 | GREGORIO MORENO-FLORES        |
| 94  | 5127444 | GREGORY SHINAULT              |
| 95  | 4272783 | GUANHUA CHEN                  |
| 96  | 4436002 | GUANNAN SUN                   |
| 97  | 1505756 | GUILHERME ROSA                |
| 98  | 4886028 | GUILHERME VIEIRA NUNES LUDWIG |
| 99  | 4884161 | GUN WOONG PARK                |
| 100 | 5010171 | HAN CHEN                      |
| 101 | 5514514 | HAO CHEN                      |
| 102 | 5520701 | HAO TENG                      |
| 103 | 4435790 | HAO ZHENG                     |
| 104 | 5323690 | HAO ZHOU                      |
| 105 | 3456935 | HAODA FU                      |
| 106 | 4725072 | HAOYANG FAN                   |
| 107 | 4149843 | HEATHER MARIE BRAZEAU         |
| 108 | 3623432 | HUAIBAO FENG                  |
| 109 | 3483534 | HUI WANG                      |
| 110 | 5168069 | HUIKUN ZHANG                  |
| 111 | 5167990 | HUILIN HU                     |
| 112 | 5321537 | HYEBIN SONG                   |
| 113 | 4734607 | HYUNSEUNG KANG                |
| 114 | 801445  | IAN BRANSTAD RILEY            |
| 115 | 143932  | ISMOR FISCHER                 |
| 116 | 5515948 | JAMES ANDERSON                |
| 117 | 2601633 | JAMES D KUELS                 |
| 118 | 4636315 | JARED BROWN                   |
| 119 | 3011936 | JASON P FINE                  |
| 120 | 3956063 | JASON RICHARD SWANSON         |
| 121 | 4140397 | JEE YEON KIM                  |
| 122 | 4890283 | JEEA CHOI                     |
| 123 | 4636145 | JENNIFER BIRSTLER             |
| 124 | 4613728 | JENNIFER NGUYEN               |
| 125 | 3627934 | JESSE THOMAS HOLZER           |
| 126 | 4590715 | JIAJIE CHEN                   |
| 127 | 4433317 | JIALE XU                      |
| 128 | 4603165 | JIAN WU                       |
| 129 | 4886987 | JIANCHANG HU                  |
| 130 | 5410648 | JIE SONG                      |
| 131 | 4292460 | JIE WEI                       |
| 132 | 4436368 | JIE ZHANG                     |

|     |         |                              |
|-----|---------|------------------------------|
| 133 | 4734274 | JILI WANG                    |
| 134 | 3783649 | JINGCI MENG                  |
| 135 | 3973318 | JINGJIANG PENG               |
| 136 | 4902482 | JINGLAN LI                   |
| 137 | 4286103 | JIWEI ZHAO                   |
| 138 | 4880968 | JOHN DAVIS                   |
| 139 | 608997  | JOHN GILLET                  |
| 140 | 3806328 | JOHN KANE                    |
| 141 | 4896511 | JOHN WILTSHIRE-GORDON        |
| 142 | 1634418 | JONATHON PETERSON            |
| 143 | 1126199 | JOSEF GINEBRA                |
| 144 | 4610245 | JOSEPH DEUTSCH               |
| 145 | 4731731 | JU HEE CHO                   |
| 146 | 3981051 | JUN LI                       |
| 147 | 1079018 | JUN SHAO                     |
| 148 | 3620114 | JUN YIN                      |
| 149 | 3781764 | JUN ZHANG                    |
| 150 | 321374  | JUN ZHU                      |
| 151 | 3095399 | JUNGWON MUN                  |
| 152 | 3620552 | JUNHEE HAN                   |
| 153 | 4541633 | JUNHO LEE                    |
| 154 | 5012817 | KAILEI CHEN                  |
| 155 | 2602032 | KAM-WAH TSUI                 |
| 156 | 4541610 | KARL BROMAN                  |
| 157 | 4929500 | KARL ROHE                    |
| 158 | 4491860 | KATHERINE GOODE              |
| 159 | 3777992 | KAZUHIKO SHINKI              |
| 160 | 4261114 | KEEGAN KORTHAUER             |
| 161 | 3127329 | KEN ONO                      |
| 162 | 3978606 | KEVIN HASEGAWA ENG           |
| 163 | 4744350 | KEVIN PACKARD                |
| 164 | 3404299 | KJELL DOKSUM                 |
| 165 | 5476019 | KRISHNAKUMAR BALASUBRAMANIAN |
| 166 | 4199106 | KRISTEN CYFFKA               |
| 167 | 5697552 | KUNLING HUANG                |
| 168 | 5317373 | KYLE HEBERT                  |
| 169 | 5478075 | KYLE HERBET                  |
| 170 | 207204  | KYUNGMAH KIM                 |
| 171 | 4435446 | LAM HO                       |
| 172 | 5521838 | LAN LUO                      |
| 173 | 3103459 | LANCINE KONATE               |
| 174 | 3443667 | LEI XU                       |
| 175 | 3437432 | LEV BORISOV                  |

|     |         |                         |
|-----|---------|-------------------------|
| 176 | 4845798 | LIAM JOHNSTON           |
| 177 | 4277002 | LIE XIONG               |
| 178 | 3980675 | LILI LAN                |
| 179 | 5693114 | LILI ZHENG              |
| 180 | 4578330 | LILUN DU                |
| 181 | 5514932 | LIN QI                  |
| 182 | 5015113 | LONG PHAN               |
| 183 | 4726678 | LU MAO                  |
| 184 | 5010560 | LU YANG                 |
| 185 | 5017774 | LUWAN ZHANG             |
| 186 | 5192960 | LUXI CAO                |
| 187 | 5302154 | MANJUSHA KANCHARLA      |
| 188 | 2601674 | MARI PALTA              |
| 189 | 4195000 | MARIA KAMENETSKY        |
| 190 | 5167906 | MARIA ROJO              |
| 191 | 245914  | MARIAN R FISHER         |
| 192 | 333334  | MARY LINDSTROM          |
| 193 | 3467530 | MATTHEW BALLARD         |
| 194 | 4280220 | MENG SONG               |
| 195 | 1114818 | MICHAEL FERRIS          |
| 196 | 783769  | MICHAEL GEORGE ILTIS    |
| 197 | 270258  | MICHAEL HOGAN           |
| 198 | 5708167 | MICHAEL KUTZLER         |
| 199 | 5840930 | MICHAEL LIOU            |
| 200 | 2601916 | MICHAEL NEWTON          |
| 201 | 989414  | MICHAEL RENE KOSOROK    |
| 202 | 394810  | MICHELLE HARRIS         |
| 203 | 6029366 | MIHAELA IFRIM           |
| 204 | 3575433 | MIN JUNG LEE            |
| 205 | 3965653 | MIN NIU                 |
| 206 | 4133622 | MING XIE                |
| 207 | 3077702 | MING YUAN               |
| 208 | 4392906 | MINJING TAO             |
| 209 | 4947463 | MITCHELL PAUKNER        |
| 210 | 3404298 | MOO K CHUNG             |
| 211 | 5514561 | MUHONG GAO              |
| 212 | 2601621 | MURRAY CLAYTON          |
| 213 | 1505727 | NATALIA DE LEON GATTI   |
| 214 | 232828  | NELLIE LAUGHLIN         |
| 215 | 3523865 | NICHOLAS HENDERSON      |
| 216 | 640407  | NICHOLAS STEPHEN KEULER |
| 217 | 5523214 | NING FAN                |
| 218 | 3882760 | NORBERT BINKIEWICZ      |

|     |         |                     |
|-----|---------|---------------------|
| 219 | 2600689 | NORMAN DRAPER       |
| 220 | 2601093 | PAUL M TERWILLIGER  |
| 221 | 967335  | PAUL RATHOUZ        |
| 222 | 5631398 | PAUL SAVARIAPPAN    |
| 223 | 5326263 | PEIGEN ZHOU         |
| 224 | 4149787 | PERLA REYES         |
| 225 | 1515839 | PHILIP WOOD         |
| 226 | 4633355 | QI JIANG            |
| 227 | 3974100 | QI TANG             |
| 228 | 3338237 | QIAN ZHIGUANG       |
| 229 | 5718599 | QING LI             |
| 230 | 4460098 | QIONG ZHANG         |
| 231 | 4582107 | QIURONG CUI         |
| 232 | 4432994 | QUEFENG LI          |
| 233 | 4161984 | QUOC TRAN           |
| 234 | 789363  | REBECCA KOSCIK      |
| 235 | 5488796 | REBECCA POST        |
| 236 | 2600141 | RICHARD A BRUALDI   |
| 237 | 2601613 | RICHARD A. JOHNSON  |
| 238 | 2601926 | RICHARD J. CHAPPELL |
| 239 | 4957895 | ROBERT HARRON       |
| 240 | 2601512 | ROBERT R MEYER      |
| 241 | 2601091 | ROBERT WARDROP      |
| 242 | 3780010 | ROBERT WAYNE GREEN  |
| 243 | 893827  | RONALD GANGNON      |
| 244 | 5887858 | RONGJUN ZHU         |
| 245 | 5690182 | RUI CHEN            |
| 246 | 4128750 | RUI TANG            |
| 247 | 3987897 | RUIFANG SONG        |
| 248 | 3343055 | RUIFENG XU          |
| 249 | 5898424 | RUNGANG HAN         |
| 250 | 5020061 | RUOSI GUO           |
| 251 | 3727371 | RYAN ZEA            |
| 252 | 4752238 | SAMUEL STECHMANN    |
| 253 | 3981781 | SANGBUM CHOI        |
| 254 | 5354849 | SCOTT HOTTOVY       |
| 255 | 3253489 | SCOTT JOSEPH HETZEL |
| 256 | 5068296 | SEAN KENT           |
| 257 | 5110390 | SEBASTIEN ROCH      |
| 258 | 4733192 | SEHO PARK           |
| 259 | 5520811 | SEULKEE YUN         |
| 260 | 3573852 | SEUNGBONG HAN       |
| 261 | 5905014 | SHAN LU             |

|     |         |                              |
|-----|---------|------------------------------|
| 262 | 982957  | SHANE HUBLER                 |
| 263 | 5515201 | SHANG WU                     |
| 264 | 4146030 | SHENG WANG                   |
| 265 | 5906594 | SHENG WANG                   |
| 266 | 4285558 | SHENG ZHANG                  |
| 267 | 5017571 | SHENGJI JIA                  |
| 268 | 3127330 | SHI JIN                      |
| 269 | 4885477 | SHIXUE LIU                   |
| 270 | 4598772 | SHIZHEN WANG                 |
| 271 | 4699427 | SHUAI CHEN                   |
| 272 | 4271924 | SHUANG HUANG                 |
| 273 | 5170885 | SHULEI WANG                  |
| 274 | 4588354 | SHUYUN YE                    |
| 275 | 3622809 | SIJIAN WANG                  |
| 276 | 5323628 | SIJING LI                    |
| 277 | 5176905 | SOHEIL SADEGHI               |
| 278 | 4593965 | SOKOL VAKO                   |
| 279 | 3955531 | SONA Z SWANSON               |
| 280 | 4953246 | SONG WANG                    |
| 281 | 3884290 | SRIKANTHMADHAVAN ARAVAMUTHAN |
| 282 | 2600973 | STEFFEN LEMPP                |
| 283 | 3122658 | STEPHEN AARON STANHOPE       |
| 284 | 4538540 | STEPHEN BERG                 |
| 285 | 2601051 | STEPHEN WAINGER              |
| 286 | 3382451 | STEPHEN WRIGHT               |
| 287 | 5452538 | STEVEN SAM                   |
| 288 | 3117835 | SUKHENDU MEHROTRA            |
| 289 | 3801422 | SUNDUZ KELES                 |
| 290 | 4590738 | TAERI UHM                    |
| 291 | 3781422 | TAO YU                       |
| 292 | 5023414 | THEVAASIINEN CHANDERENG      |
| 293 | 417071  | THOMAS COOK                  |
| 294 | 2601039 | THOMAS G. KURTZ              |
| 295 | 4904255 | THU LE                       |
| 296 | 4896939 | TIEN VO                      |
| 297 | 3374880 | TIMO SEPPALAINEN             |
| 298 | 4879129 | TIMOTHY IDOWU                |
| 299 | 4957916 | TIMOTHY IDOWU                |
| 300 | 5520995 | TING YE                      |
| 301 | 3351080 | TING-LI LIN                  |
| 302 | 5329574 | TONG LI                      |
| 303 | 3230527 | TONGHAI YANG                 |
| 304 | 4896172 | TRAM TA                      |

|     |         |                    |
|-----|---------|--------------------|
| 305 | 5567363 | TUN LEE NG         |
| 306 | 5664494 | TZU HSIANG HUNG    |
| 307 | 3959251 | VARSHA KULKARNI    |
| 308 | 4641200 | VICTOR LUO         |
| 309 | 4788841 | VICTORIA MANSFIELD |
| 310 | 4434130 | WAI TONG FAN       |
| 311 | 3780817 | WEI ZHENG          |
| 312 | 2602108 | WEI-YIN LOH        |
| 313 | 4590481 | WENWEN ZHANG       |
| 314 | 5322094 | WENZHI CAO         |
| 315 | 5161294 | WESLEY CHANG       |
| 316 | 4591139 | XIAO GUO           |
| 317 | 4905505 | XIAO NIE           |
| 318 | 3343005 | XIAODAN WEI        |
| 319 | 5170801 | XIAOMAO LI         |
| 320 | 4729481 | XIAOPING FENG      |
| 321 | 5258581 | XIAOWU DAI         |
| 322 | 4285528 | XIN LI             |
| 323 | 4588103 | XIN ZHANG          |
| 324 | 4726666 | XINJIE HE          |
| 325 | 3620612 | XINWEI DENG        |
| 326 | 4286679 | XINXIN YU          |
| 327 | 4430246 | XINYU SONG         |
| 328 | 4145889 | XIPEI YANG         |
| 329 | 5170924 | XIRAN WANG         |
| 330 | 4194670 | XIUFENG SHAO       |
| 331 | 5322566 | XIUYU MA           |
| 332 | 3791802 | XIWEN MA           |
| 333 | 4286138 | XU HE              |
| 334 | 4590559 | XU XU              |
| 335 | 4726703 | XUEYAO CHEN        |
| 336 | 3668575 | XUN ZHAO           |
| 337 | 4426932 | YAJUAN SI          |
| 338 | 3800708 | YALI WANG          |
| 339 | 4721620 | YAN CHEN           |
| 340 | 3457484 | YANBING ZHENG      |
| 341 | 4285354 | YANG ZHAO          |
| 342 | 4172260 | YANNAN QIU         |
| 343 | 4895244 | YAOGUO XIE         |
| 344 | 4576059 | YAoyao XU          |
| 345 | 4695492 | YAZHEN WANG        |
| 346 | 4285538 | YI CHAI            |
| 347 | 5327139 | YI LI              |

|     |         |                 |
|-----|---------|-----------------|
| 348 | 4586128 | YI LIU          |
| 349 | 5693277 | YIFAN MEI       |
| 350 | 5323691 | YILIN ZHANG     |
| 351 | 5170492 | YING ZHANG      |
| 352 | 5696436 | YONGFENG WU     |
| 353 | 5644228 | YONGJOON KIM    |
| 354 | 5518378 | YONGSU LEE      |
| 355 | 5514517 | YOUNG LEE       |
| 356 | 5177493 | YOUNG MIN PARK  |
| 357 | 4278664 | YOUNGDEOK HWANG |
| 358 | 5326059 | YOURAN QI       |
| 359 | 3010828 | YU MENGANG      |
| 360 | 3961448 | YU QIU LIU      |
| 361 | 3791787 | YUAN JIANG      |
| 362 | 5167973 | YUAN LI         |
| 363 | 4733966 | YUAN WANG       |
| 364 | 4906458 | YUANZHI LI      |
| 365 | 5903922 | YUCHANG WU      |
| 366 | 5700058 | YUCHEN ZHOU     |
| 367 | 4144479 | YUJIN CHUNG     |
| 368 | 5907599 | YUNONG LIN      |
| 369 | 5214858 | YUQING XU       |
| 370 | 1426304 | ZHANG CHUNMING  |
| 371 | 4108525 | ZHENGJUN ZHANG  |
| 372 | 3107260 | ZHENGXIAO WU    |
| 373 | 4588525 | ZHIGENG GENG    |
| 374 | 3339239 | ZHIGUO XIAO     |
| 375 | 5693629 | ZHONGJIE YU     |
| 376 | 3626007 | ZHUANG WU       |
| 377 | 5099221 | ZIFENG ZHAO     |
| 378 | 5591917 | ZIJIAN NI       |
| 379 | 4138542 | ZUOFENG SHANG   |

3)

```
query_gpa_professores <- "
  SELECT i.name,
         SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 +
             gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count * 1.0) * 1.0 /
         SUM(gd.a_count + gd.ab_count + gd.b_count +
             gd.bc_count + gd.c_count + gd.d_count + gd.f_count) AS avg_gpa
  FROM instructors i
  JOIN teachings t ON i.id = t.instructor_id
```

```

JOIN sections sec ON t.section_uuid = sec.uuid
JOIN course_offerings co ON sec.course_offering_uuid = co.uuid
JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects s ON sm.subject_code = s.code
WHERE s.abbreviation = 'STAT'
GROUP BY i.id, i.name
HAVING SUM(gd.a_count + gd.ab_count + gd.b_count +
           gd.bc_count + gd.c_count + gd.d_count + gd.f_count) > 0
ORDER BY avg_gpa ASC
"

gpa_professores <- dbGetQuery(conn, query_gpa_professores)

prof_mais_dificil <- head(gpa_professores, 1)
prof_mais_facil <- tail(gpa_professores, 1)

cat("Professor mais difícil (menor GPA ponderado):\n")

```

Professor mais difícil (menor GPA ponderado):

```
print(prof_mais_dificil)
```

```

      name avg_gpa
1 JAMES D KUELBS 2.607692

```

```
cat("\nProfessor mais fácil (maior GPA ponderado):\n")
```

Professor mais fácil (maior GPA ponderado):

```
print(prof_mais_facil)
```

```

      name avg_gpa
374 YAJUAN SI      4

```

```

query_gpa_disciplinas <- "
  SELECT c.number, c.name,
         SUM(gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 +
             gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count * 1.0) * 1.0 /
         SUM(gd.a_count + gd.ab_count + gd.b_count +
             gd.bc_count + gd.c_count + gd.d_count + gd.f_count) AS avg_gpa
  FROM courses c
  JOIN course_offerings co ON c.uuid = co.course_uuid
  JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
  JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
  JOIN subjects s ON sm.subject_code = s.code
  WHERE s.abbreviation = 'STAT'
  GROUP BY c.uuid, c.name
  HAVING SUM(gd.a_count + gd.ab_count + gd.b_count +
             gd.bc_count + gd.c_count + gd.d_count + gd.f_count) > 0
  ORDER BY avg_gpa ASC
"

gpa_disciplinas <- dbGetQuery(conn, query_gpa_disciplinas)

disciplina_mais_dificil <- head(gpa_disciplinas, 1)
disciplina_mais_facil   <- tail(gpa_disciplinas, 1)

cat("\nDisciplina mais difícil (menor GPA ponderado):\n")

```

Disciplina mais difícil (menor GPA ponderado):

```
print(disciplina_mais_dificil)
```

|   | number | name                                      | avg_gpa  |
|---|--------|-------------------------------------------|----------|
| 1 | 431    | Introduction to the Theory of Probability | 2.886374 |

```
cat("\nDisciplina mais fácil (maior GPA ponderado):\n")
```

Disciplina mais fácil (maior GPA ponderado):

```
print(disciplina_mais_facil)
```

```
      number              name avg_gpa  
70      834 Empir Proc&Semiparmtrc Infernc      4
```

```
4)
```

```
dbDisconnect(conn)
```

```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-17 11:55:03 -03"
```